conditions (generally dry and stable) over inland northern Australia and the top end. In the summer months, the sub-tropical ridge moves south, the Trade Winds weaken and the monsoon returns to northern Australia.

72.

1. **Tropical depressions may bring heavy rainfall** to affected parts of northern Australia. Tropical depressions are moderate-strength low pressure systems which occur in the tropics, often in association with the monsoon trough.

   How do they affect Australia?

   Tropical depressions are associated with occasional thunderstorms, fresh to strong and gusty winds and heavy rain, often leading to flooding of affected areas.
2. Tropical cyclones affect all of northern Australia, with the northwest of Western Australia between Broome and Exmouth being the most cyclone prone part of Australia's coastline.

   The Australian cyclone season officially runs from November to April, although very few have occurred in November.

73. Upper level troughs can bring enhanced rainfall to Australia.

What are they?

An upper level trough is a trough of low pressure which has formed in the upper levels of the atmosphere, and hence cannot be seen on surface level charts.

Upper level troughs can occur at any time, in any region of Australia. They generally last from between a few days to a couple of weeks and typically move from west to east.

How do they affect Australia?

Upper level troughs can result in the formation of a cloudbands, which will often result in widespread rainfall near and to the east of the trough. Upper level troughs can also aid in the development of surface level features, such as frontal systems, thereby enhancing their effect. The presence of an upper level trough in a favourable position may result in the development of a cut-off low, which in turn will enhance rainfall over the affected region.

74. **The Indonesian Throughflow**

As the name suggests the Indonesian Throughflow is a system of currents that carries water westward from the Pacific to the Indian